NUST Historical Batch Processing — Cost Estimate

Date: 2026-04-27 **Baseline:** 1980 processing = $13.50 (164 entries, 10 tests, 2 XLSX files)

Cost Drivers

Each XLSX file is chunked by group (tp2 markers); large groups split further at tp6/tp7. Every chunk = 1+ Claude API calls (`claude-sonnet-4-6`). Total cost scales with:

- Number of entries (strains) per year
- Number of tests (maturity groups) per year
- Prompt length (era-aware addenda add input tokens)

Estimate by Era

Era	Years	Avg entries	Avg tests	Est. $/yr	Subtotal
Early (1941–1955)	~15	30–60	3–5	$3–5	~$60
Mid (1956–1969)	~14	60–100	4–6	$5–8	~$90
Transition (1970–1979)	~9	100–140	6–8	$8–12	~$90
Late (1981–1986)	~6	~160	8–10	~$13	~$80

Estimated total: $300–420 for ~44 XLSX-processable years (~$7–9/year weighted average).

Exclusions

Year(s)	Reason	Status
1980	Already processed	Complete
1975	PDF-only, no XLSX	Separate extraction path needed
1987–1988	Fragmented XLSX structure	Separate assembler needed

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Caveats (could push cost higher)

- **OCR artifacts** in early years may trigger multi-turn clarification calls
- **Era-aware system prompt** addenda increase input token count per call
- **1987/1988 fragmented files** may require multiple re-processing passes
- **1975 PDF path** cost unknown until that pipeline is built

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Recommended Approach: Staged Batch Run

Before committing to the full run, sample one year from each era to calibrate actual per-call cost:

1. Pick a year from 1941–1955 (e.g., 1950)
2. Pick a year from 1956–1969 (e.g., 1963)
3. Pick a year from 1970–1979 (e.g., 1972)

~$30 total to validate all three eras before spending the remainder.

Reference

- **Extraction script:** `scripts/extract_nust_xlsx.py`
- **Validation script:** `scripts/validate_nust_hist.py`
- **Era-aware prompt notes:** `docs/system_prompt_multiyr_notes.md`
- **Source XLSX files:** `R:/NUST_Historical_Data/Green-*/Green/`